

Final Project Report

Biological Question and Context

The biological question for this project is: How do tau pathology and aging alter cell-type-specific hippocampal gene expression and spatial gene-expression patterns in an Alzheimer's disease tauopathy rat model? This question focuses on whether Tau transgenic rats show transcriptional changes in specific hippocampal cell populations and whether these changes differ between 10-month and 20-month animals. This revised question better matches the analyses performed in my current script, which focuses on single-nucleus RNA-seq clustering, cell-type annotation, cell-type proportion analysis, differential expression, and spatial transcriptomics.

This question is biologically relevant because Alzheimer's disease involves progressive changes in neuronal function, synaptic signaling, glial responses, and brain-region vulnerability. The paper associated with this dataset uses the R955-hTau transgenic rat model, which overexpresses human tau with the P301S mutation, to study tau-driven neurodegenerative mechanisms in the hippocampus. The authors specifically focused on early 10-month and later 20-month pathology in Tau rats and age-matched wild-type controls.

Transcriptomics is appropriate for this question because tau pathology is unlikely to affect only one gene or one cell type. Instead, it may alter broad gene-expression programs across neurons, astrocytes, microglia, oligodendrocytes, OPCs, vascular-associated cells, and other hippocampal populations. Single-nucleus RNA sequencing is useful because it allows these changes to be analyzed at cell-type resolution rather than averaging all cells together. This is especially important in brain tissue, where changes in a small but biologically important cell population could be missed by bulk RNA-seq.

The hippocampus is an appropriate tissue focus because it is central to learning and memory and is vulnerable in Alzheimer's disease. The linked study emphasizes that tau

pathology and aging can alter synaptic organization, glutamatergic signaling, neurotrophic signaling, and other molecular pathways relevant to hippocampal function. Therefore, this project uses transcriptomics to explore whether Tau rats show altered hippocampal gene expression at the level of specific cell types and spatial tissue regions.

Dataset Identification and Experimental Design

The dataset selected for this project is GSE305314 from the NCBI Gene Expression Omnibus. The dataset is a single-cell multiome and spatial transcriptomics study of a transgenic rat model of human-like tauopathy. The dataset is associated with the paper *Intercellular signaling and synaptic deconstruction uncovered by single-cell and spatial transcriptomics in an AD tauopathy model* (Ji et al. 2025).

The full GEO dataset contains 28 samples, including snRNA-seq, scATAC-seq, and Visium spatial transcriptomics samples. For this project, the current analysis uses the 12 snRNA-seq samples and the 4 Visium spatial transcriptomics samples. The snRNA-seq samples include three Tau 10-month samples, three WT 10-month samples, three Tau 20-month samples, and three WT 20-month samples. The spatial dataset includes one Visium sample for each of the four experimental groups: Tau 10 mo, WT 10 mo, Tau 20 mo, and WT 20 mo.

The experimental design includes two independent variables: genotype and age. Genotype compares Tau transgenic rats with wild-type controls, while age compares 10-month and 20-month animals. The WT animals serve as controls for the Tau animals, and the 20-month WT group is especially important because it helps distinguish normal aging effects from tau-associated disease effects.

In the script, the main snRNA-seq workflow begins by loading the 12 samples into Seurat objects. Each sample is assigned metadata for sample identity, condition, and age. Doublets are removed using scDbtFinder, and mitochondrial percentage is calculated. The data are then filtered using $nFeature_RNA < 5000$ and $percent.mt < 10$.

These quality-control steps are important because doublets, damaged nuclei, and low-quality nuclei can distort clustering and downstream differential expression results.

After quality control, the samples are merged into a single Seurat object. The data are normalized with `NormalizeData`, variable features are identified with `FindVariableFeatures`, the data are scaled, and PCA is performed. The first 19 dimensions are used for clustering and UMAP visualization after an elbow plot analysis. Harmony integration is then applied to reduce sample-level or batch-related effects while preserving biological variation. After Harmony's batch correction, the object is reclustered and visualized with UMAP.

Cell-type annotation is performed using cluster marker genes identified by `FindAllMarkers`. The current analysis annotates major hippocampal populations, including oligodendrocytes, dentate gyrus (DG)/CA3 glutamatergic neurons, astrocytes, OPCs, CA1 glutamatergic neurons, subiculum-related glutamatergic neurons, microglia, GABAergic neuron subtypes (`Lamp5`, `Lhx6`, `Sst`, `Pvalb`, `VIP`, etc.), vascular leptomeningeal cells (VLMCs), choroidal epithelial cells (`Chor`), endothelial cells, pericytes, ependymal cells, and other immune cells. Marker genes are used to support these annotations.

Cell-type proportion analysis was performed using `dittoBarPlot` and `scProportionTest`. Cell-type proportions are compared by condition, meaning Tau vs WT, and by age, meaning 10 mo vs 20 mo. This is useful because apparent gene-expression changes can sometimes reflect shifts in cell composition rather than true transcriptional changes within a cell type.

Differential expression is performed within annotated cell types using `FindMarkers`. In the current version of the analysis, Tau vs WT comparisons are performed within each cell type, collapsed across age. Age comparisons are also performed within each cell type, collapsed across genotype.

Multi-modal Strategy

The primary modality used in this project is single-nucleus RNA sequencing. This modality provides cell-type-level resolution and allows gene-expression differences to be tested within specific hippocampal populations. This is important because tau pathology may not affect all cell types equally. For example, neurons may show changes related to synaptic signaling, while glial populations may show changes related to support functions, inflammation, or myelination.

The second modality used in this project is 10x Visium spatial transcriptomics. Spatial transcriptomics adds biological value because snRNA-seq requires dissociation or nuclei isolation, which removes cells from their anatomical context. Visium preserves tissue location, allowing gene-expression patterns to be examined across spatial regions. The dataset includes both single-cell multiome data and spatial transcriptomics from Tau and WT rat brain tissues at 10 and 20 months.

In the spatial portion of the analysis, four Visium samples are loaded: Tau 10 mo, Tau 20 mo, WT 10 mo, and WT 20 mo. Metadata are added for sample, condition, and age. The script visualizes spatial QC metrics such as `nFeature_Spatial` and `nCount_Spatial`, then normalizes each spatial object with `SCTransform`. The spatial samples are merged, and PCA, clustering, UMAP, and spatial cluster visualization are performed. This allows transcriptionally distinct spatial clusters to be mapped back onto tissue sections.

The spatial analysis also identifies spatially variable genes using Moran's I and identifies cluster marker genes using `FindAllMarkers`. Marker genes were used to help interpret spatial clusters. For example, the script uses markers to label broad regions or structures such as fiber tracts, cortical layers, thalamus, hypothalamus, olfactory areas, striatum, dentate gyrus, and CA1-like regions.

This multimodal strategy strengthens the project because the two data types answer related but different questions. The snRNA-seq analysis identifies which cell types show transcriptional differences by tau condition or age. The spatial transcriptomics analysis shows where gene-expression patterns are located in tissue. Together, these

approaches provide a more complete picture of how tau pathology and aging affect the hippocampus than either modality alone.

Data Exploration and Interpretation

The single-nucleus RNA-seq analysis began by loading the 12 snRNA-seq samples into Seurat. These samples represented four experimental groups: Tau 10 months, Tau 20 months, wild-type 10 months, and wild-type 20 months. This design matches the project plan, where genotype and age are the two independent variables and gene expression and cell–cell communication are the major dependent variables. Each sample was converted into a Seurat object, and metadata were added to identify the sample name, condition, and age. This metadata structure allowed later comparisons between Tau and wild-type animals and between 10-month and 20-month animals.

Quality control was performed before downstream analysis to reduce noise from ambient RNA contamination and technical artifacts. Doublets were identified with scDbtFinder, and only singlets were retained for analysis. Mitochondrial percentage was calculated using mitochondrial gene expression, and nuclei were filtered using two main thresholds: fewer than 5,000 detected RNA features and less than 10% mitochondrial gene expression. These filtering steps were important because low-quality nuclei, damaged nuclei, and doublets can distort clustering and differential expression results. Violin plots of nFeature_RNA, nCount_RNA, and percent.mt were used to evaluate sample quality before final filtering.

After quality control, the samples were merged into one Seurat object and normalized. Variable genes were identified, the data were scaled, and PCA was performed. Based on the elbow plot, the first 19 principal components were used for neighbor finding, clustering, and UMAP visualization. Harmony integration was then applied to reduce sample-level or batch-related variation while preserving biological differences among cell types and conditions. After Harmony correction, the data were reclustered and visualized using UMAP. The UMAP showed that cells grouped into distinct clusters rather than forming a single mixed population, suggesting that the dataset captured meaningful cellular heterogeneity in the hippocampus.

Cluster annotation was performed using marker genes identified with FindAllMarkers and visualized with FeaturePlots, DotPlots, and heatmaps. Major hippocampal cell types were identified, including oligodendrocytes, astrocytes, oligodendrocyte precursor cells, microglia, glutamatergic neurons, GABAergic neurons, vascular leptomeningeal cells, choroidal epithelial cells, pericytes, endothelial cells, and ependymal cells. Neuronal subtypes were further separated using marker genes. This annotation step was necessary because the biological question is not only whether tau pathology changes gene expression, but also which cell types show those changes.

Cell-type proportion analysis was then used to test whether tau pathology or aging altered the cellular composition of the hippocampal dataset. Bar plots were generated with dittoBarPlot to compare the relative abundance of annotated cell types across condition and age. In addition, permutation testing with scProportionTest was used to statistically evaluate whether cell-type proportions differed between Tau and wild-type animals or between 10-month and 20-month animals. This analysis is important because apparent gene-expression changes in a whole tissue dataset can sometimes reflect shifts in cell composition rather than true transcriptional changes within a cell type. By analyzing cell proportions separately, the project can better distinguish between changes in cellular abundance and changes in gene expression within specific populations.

Differential expression analysis was performed within each annotated cell type using Seurat's FindMarkers. This cell-type-specific approach is appropriate for the project because tau pathology may affect neurons, glia, and vascular-associated cells differently. One set of comparisons tests wild-type versus Tau within each cell type, while another set tests 10 months versus 20 months within each cell type. These comparisons provide an initial view of genotype-associated and age-associated transcriptional changes.

Several representative genes were visualized with FeaturePlots after differential expression testing. Genes were plotted across UMAP space and split by condition or age. These plots are useful because they show whether a candidate gene is broadly expressed across many clusters or enriched in a specific cell population.

Spatial transcriptomics was used as a second layer of data exploration. Four Visium samples were loaded: Tau 10 months, Tau 20 months, wild-type 10 months, and wild-type 20 months. These samples were normalized with SCTransform, merged, and analyzed using PCA, clustering, UMAP, spatial visualization, and CellChat for cell-cell communication. Spatial clustering allowed transcriptionally distinct regions to be mapped back onto tissue sections. This is important because snRNA-seq identifies cell types but loses anatomical position, while Visium preserves spatial information. SpatialFeaturePlots were used to visualize genes, and spatially variable genes were identified using Moran's I. This analysis helps determine whether tau-associated transcriptional changes are regionally organized rather than uniformly distributed across the tissue.

The spatial analysis also identified region- or structure-associated marker genes. These markers can be used to connect spatial clusters to broad anatomical or cell-enriched regions, including fiber tracts, cortical layers, thalamus, hypothalamus, olfactory areas, striatum, dentate gyrus, and CA1-like regions. This spatial information adds biological value because hippocampal tau pathology may affect specific subregions more strongly than others. Therefore, integrating snRNA-seq and spatial transcriptomics provides a more complete picture of tau-associated gene-expression changes.

Overall, these analyses helped to answer the biological question. The snRNA-seq analysis identifies the major hippocampal cell types and allows tau- and age-associated transcriptional changes to be tested within each cell population. The cell proportion analysis evaluates whether tau pathology or aging changes the relative abundance of cell types. The spatial transcriptomics analysis adds anatomical context by showing where gene-expression programs are enriched in tissue. Together, these results support an interpretation in which tau pathology affects the hippocampus through both cell-type-specific transcriptional changes and spatially organized molecular changes. This directly addresses the project's goal of understanding how gene expression in the hippocampus changes from early to late stages of tau pathology.

Critical Reflection

Some of the limitations in the dataset included that only female samples were used for the majority of the data, and that there were only 3 samples at each age group for each condition (12 total) for single cell then 1 sample for each age group in each condition for spatial, limiting the amount of biological variance that could've been analyzed.

A limitation of the analysis (due to time) was the lack of age-specific comparisons between wild-type and tau pathology groups; for example, Tau 10 mo vs WT 10 mo and Tau 20 mo vs WT 20 mo. These comparisons would help to further answer whether tau pathology produces different transcriptional effects at early and later stages of disease.

I ran into several technical issues. My largest issue was determining the correct resolution and dimensions for clustering and batch correction so as to not void biological variation yet group together the correct clusters based on their markers. This issue led to several days of reclustering and tuning parameters, which was ultimately the lengthiest part of this project. I tried multiple different techniques to help with clustering, like DoubletFinder and SoupX (was discussed in class, I attempted to use it but it removed too many cells and used up too much memory). As the deadline was nearing, I decided to keep my best parameters even though there remained ambiguous clusters. Some other technical limitations faced during this project were the vector memory limits, which obstructed me from running more accurate and biologically helpful analyses like SCTransform on both the single-cell and spatial data. Due to too much data in the single-cell datasets, I was unable to run SCTransform and had to use log normalization, which reduced the biological variation detected in clustering. By looking at the paper, I also noticed they used techniques like pseudobulking and CellChat for both spatial and single cell, which I was unable to do due to not knowing how to do pseudobulking effectively for performing CellChat on single cell data.

Images

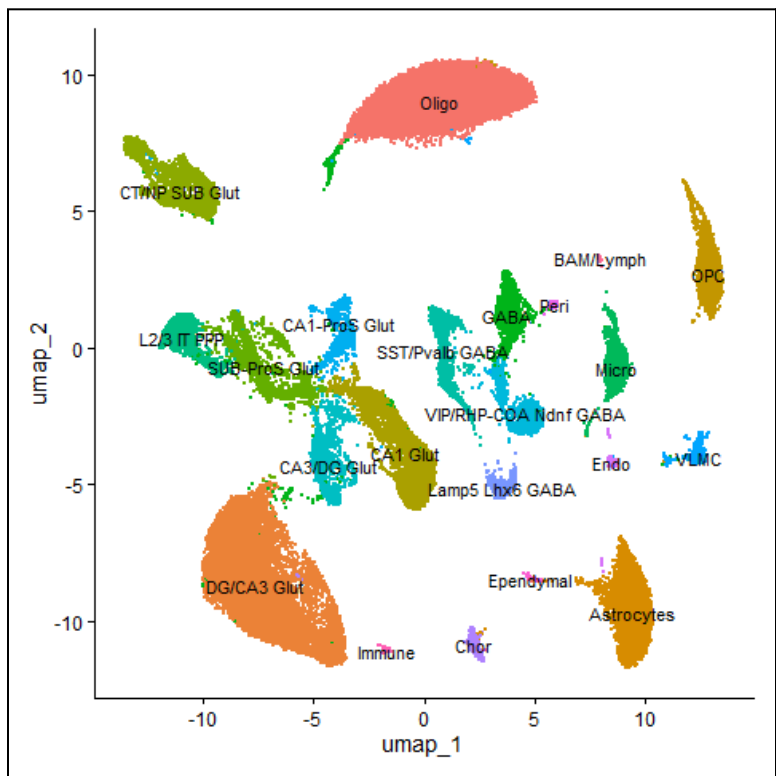


Fig. 1: UMAP plot of annotated cell types for single cell analysis

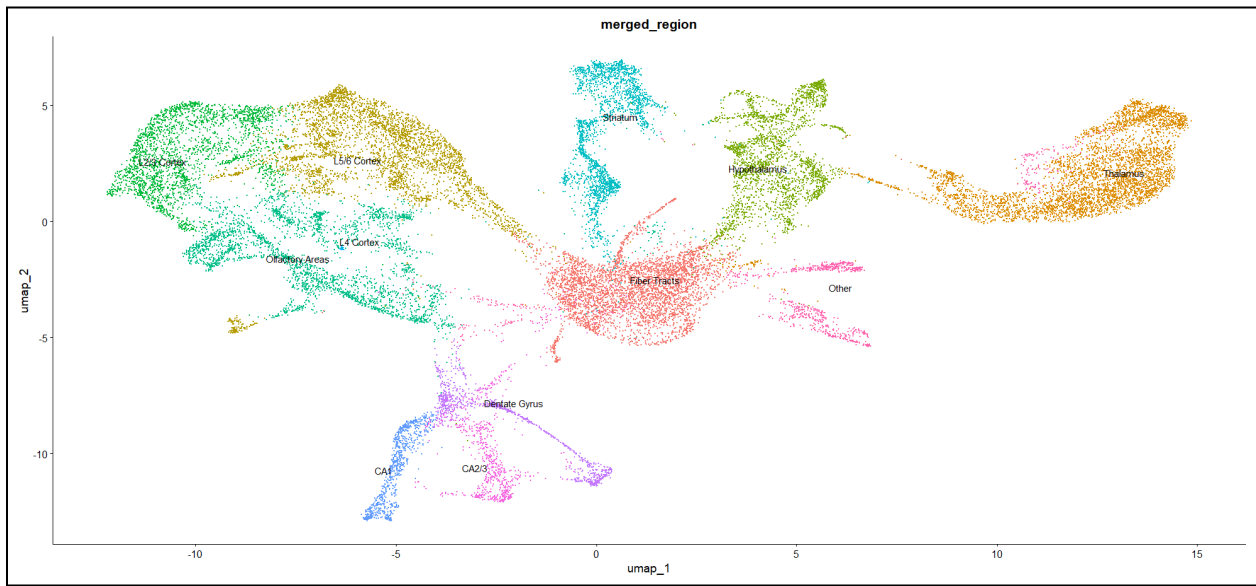


Fig. 2: UMAP plot of annotated cell types for spatial transcriptomics analysis

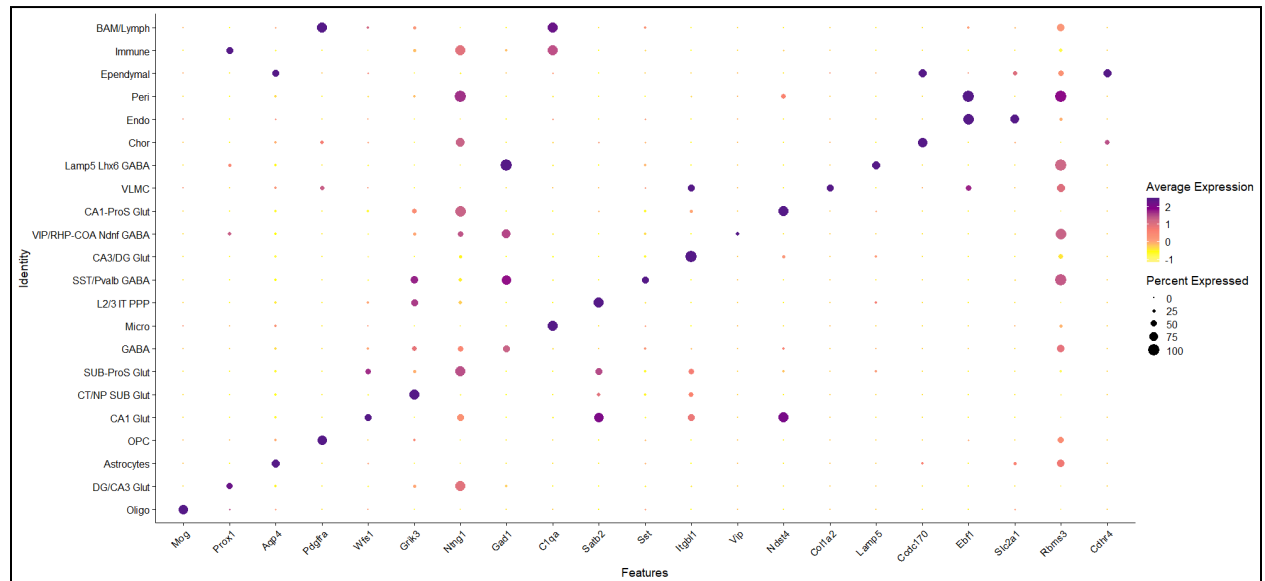


Fig. 3: Dot plot of gene markers for each annotated cluster in single cell analysis

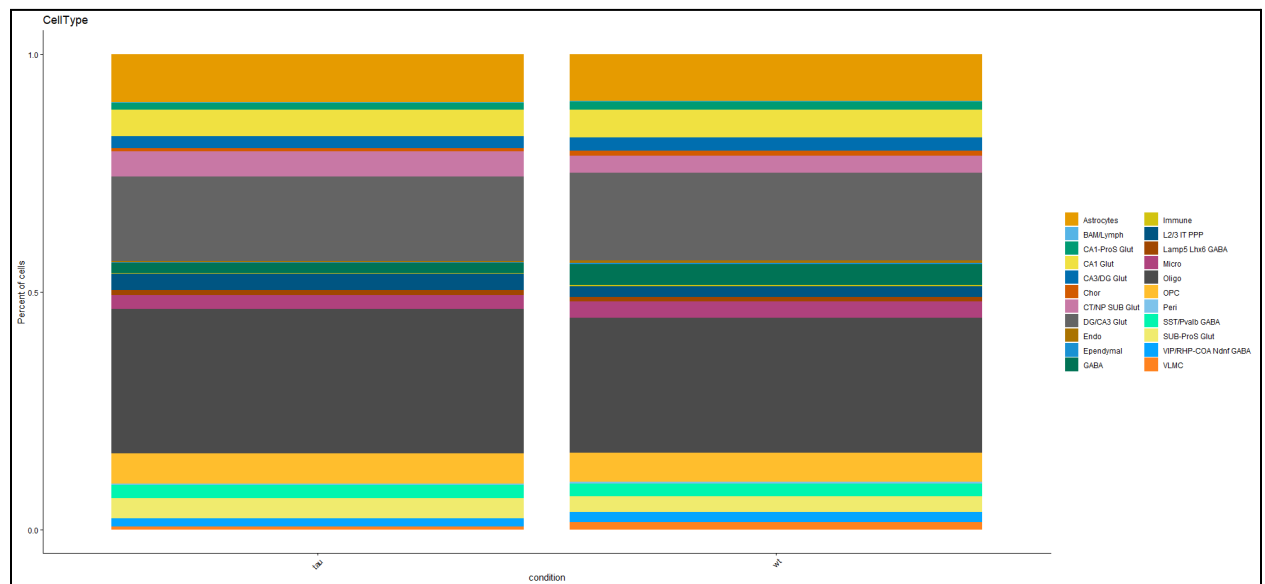


Fig. 4: Proportion of cell types across conditions in single-cell analysis

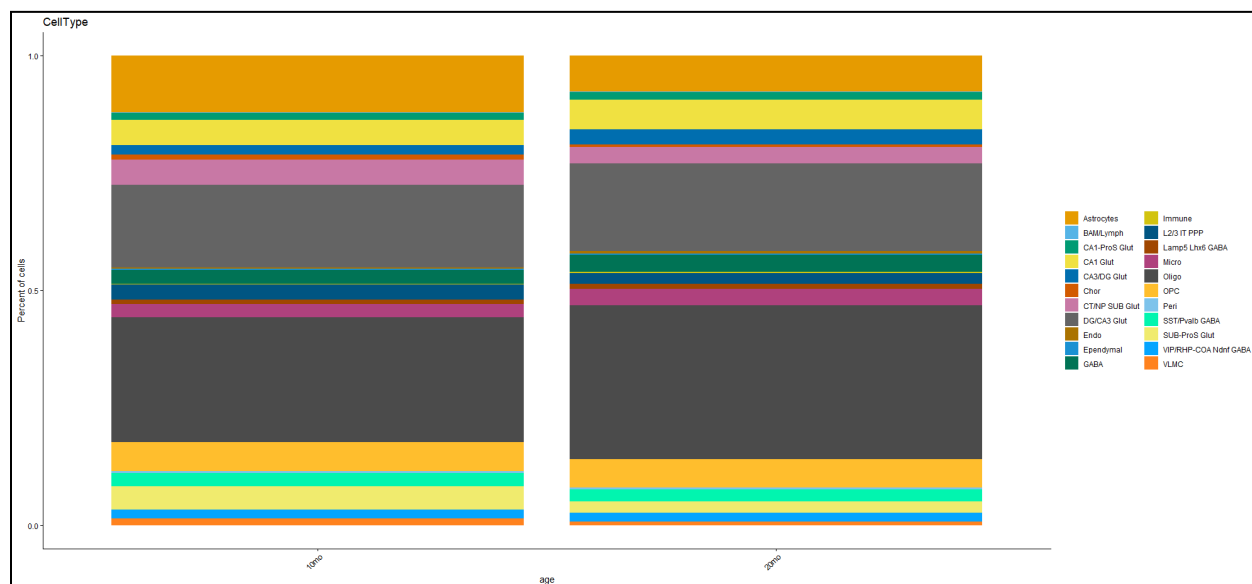


Fig. 5: Proportion of cell types across age groups in single-cell analysis

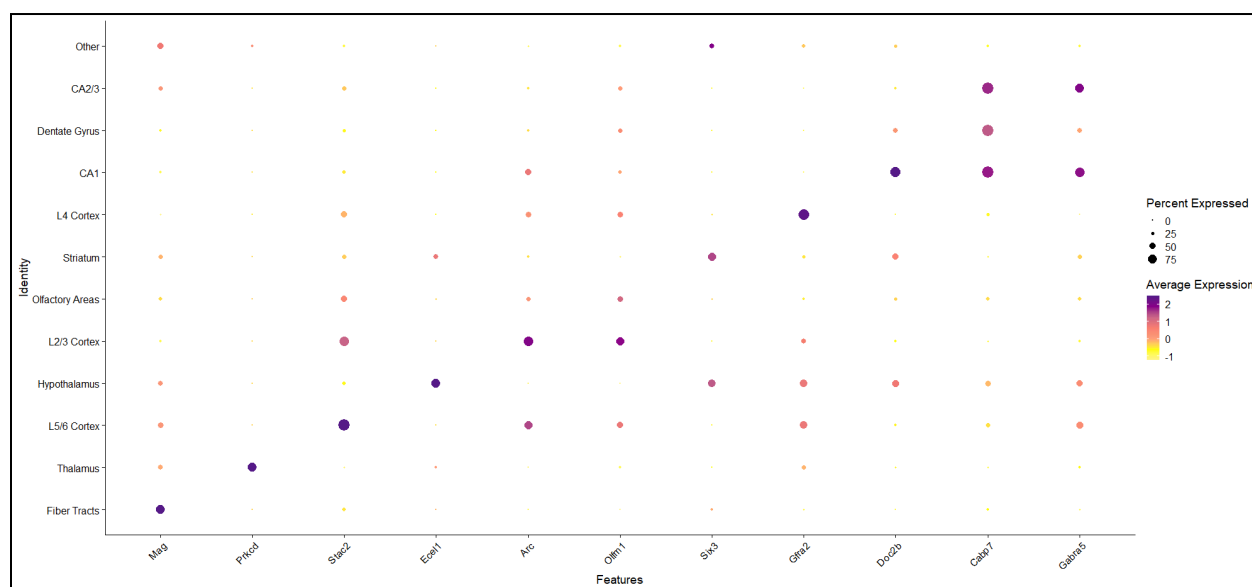


Fig. 6: Annotated regions for spatial transcriptomics against each region's marker genes

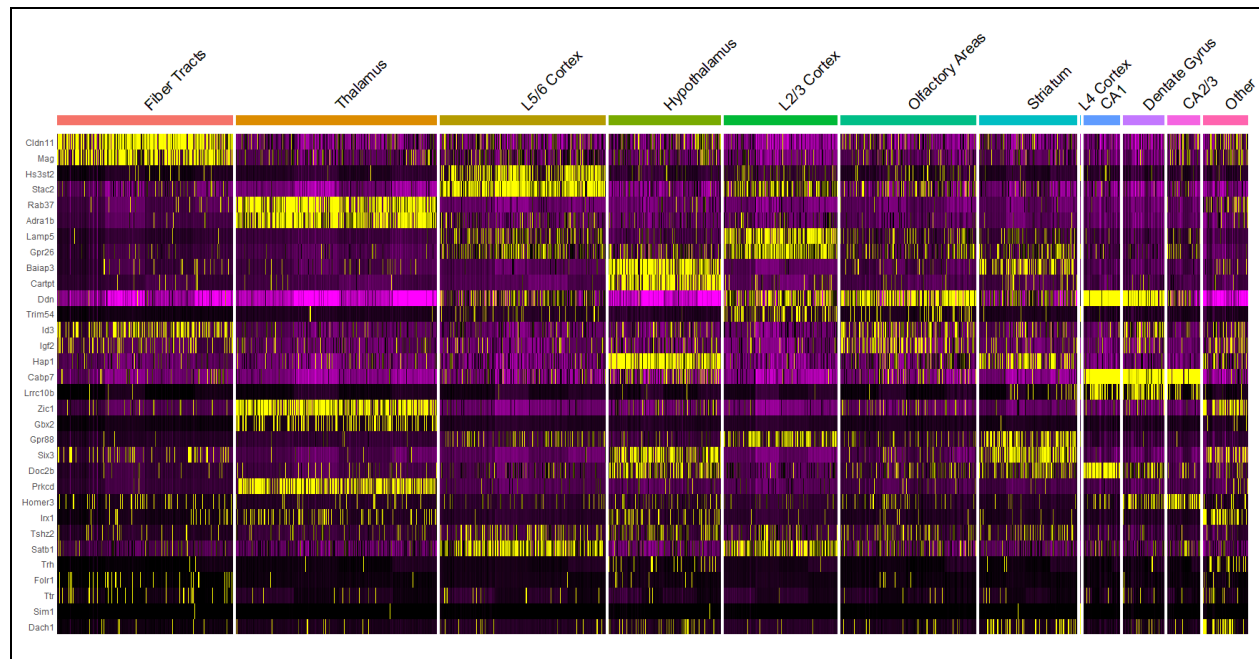


Fig. 7: Heatmap of top 2 features for each region in spatial transcriptomics

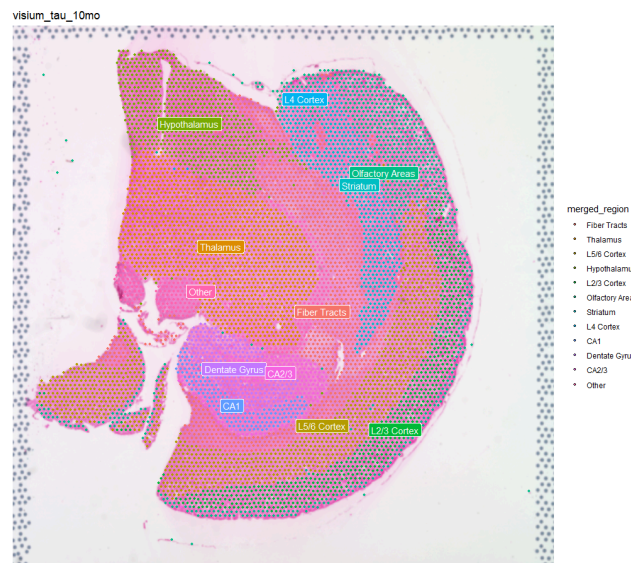


Fig. 8: Annotated clusters on Tau 10mo

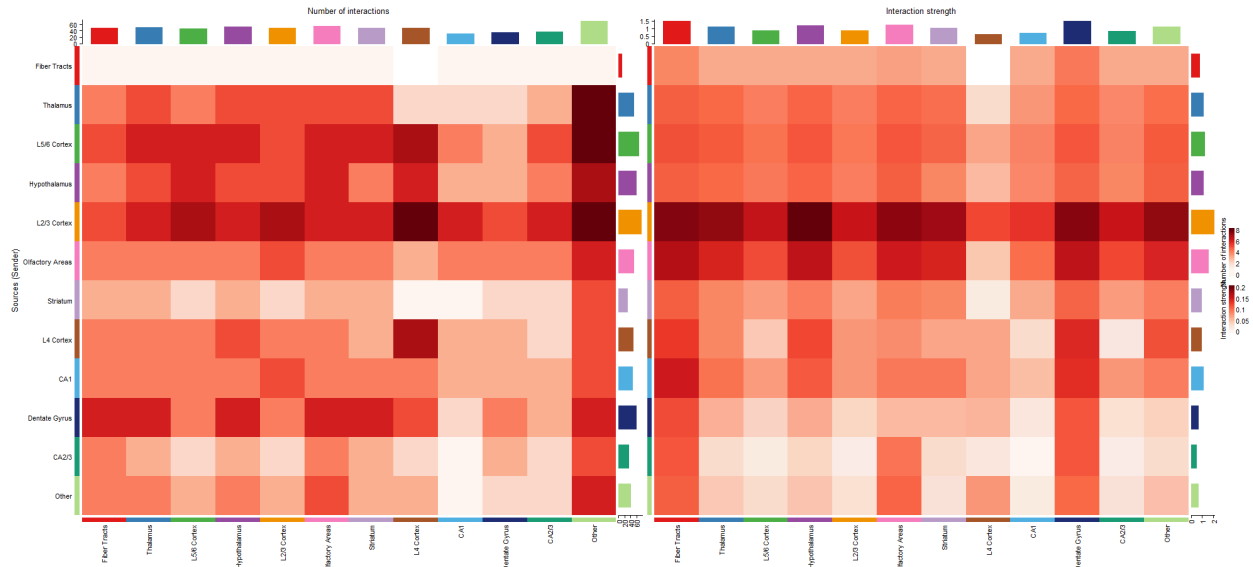


Fig. 9: Heatmap of cell-cell communication after processing Tau 10mo through CellChat

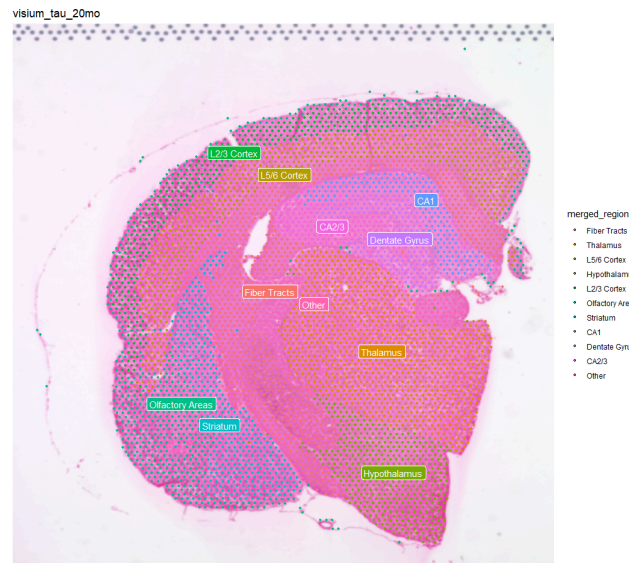


Fig. 10: Annotated clusters on Tau 20mo

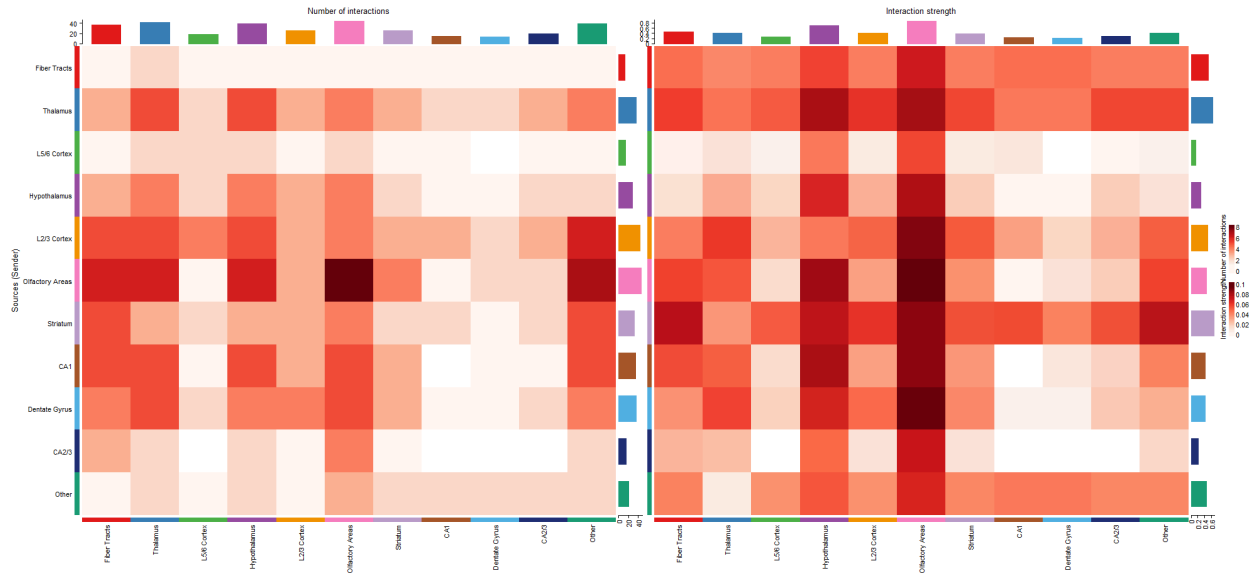


Fig. 11: Heatmap of cell-cell communication after processing Tau 20mo through CellChat

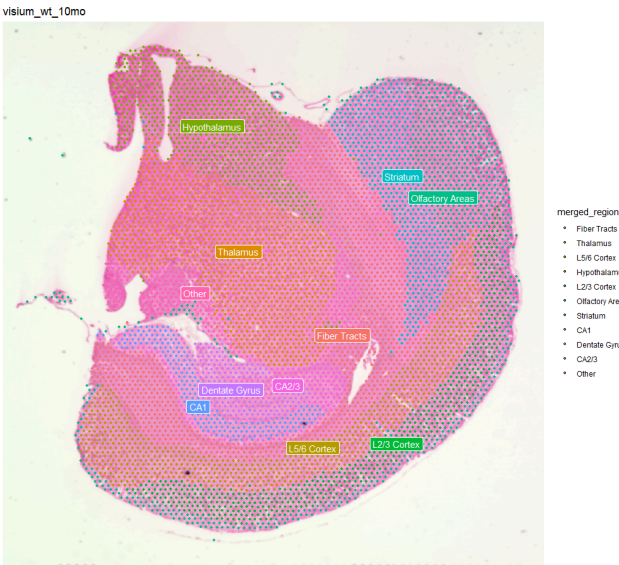


Fig. 12: Annotated clusters on WT 10mo

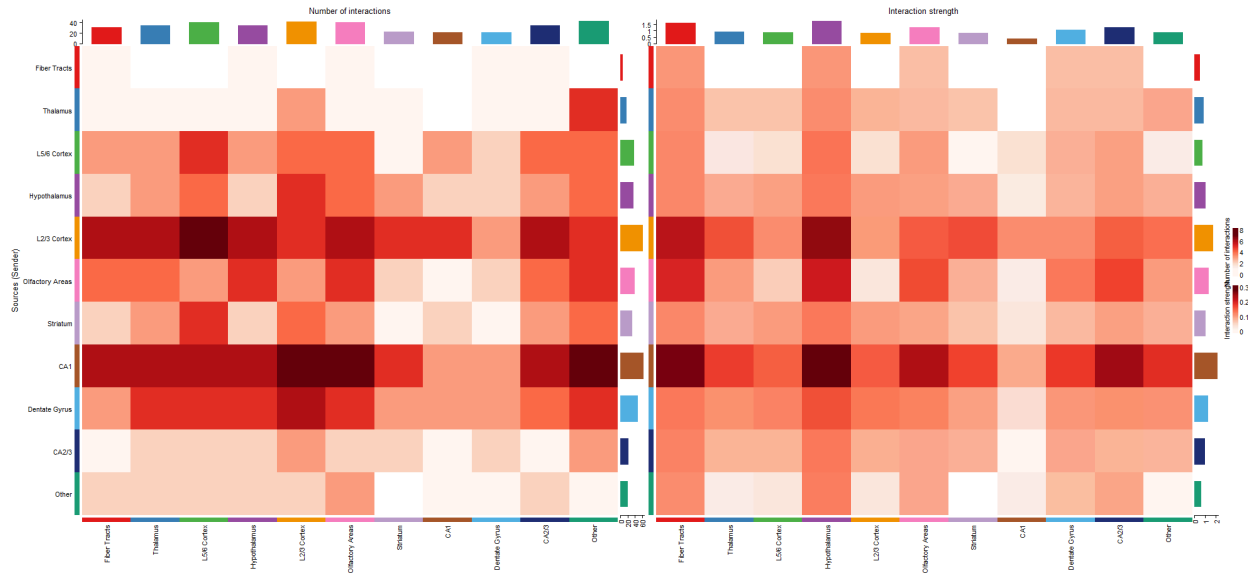


Fig. 13: Heatmap of cell-cell communication after processing WT 10mo through CellChat

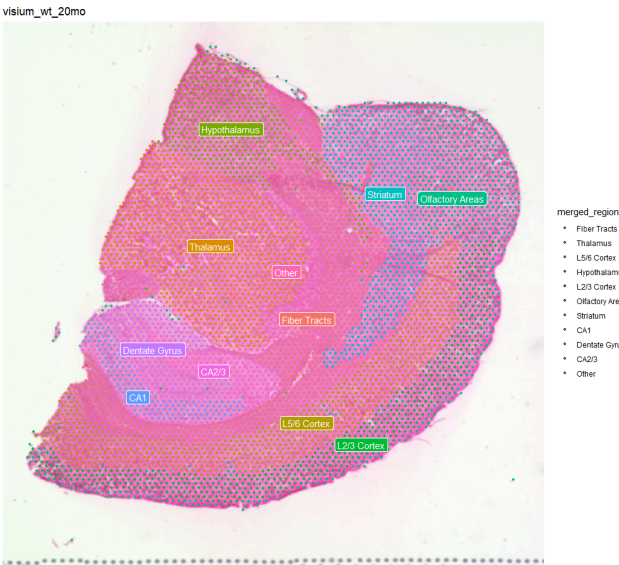


Fig. 14: Annotated clusters on WT 20mo

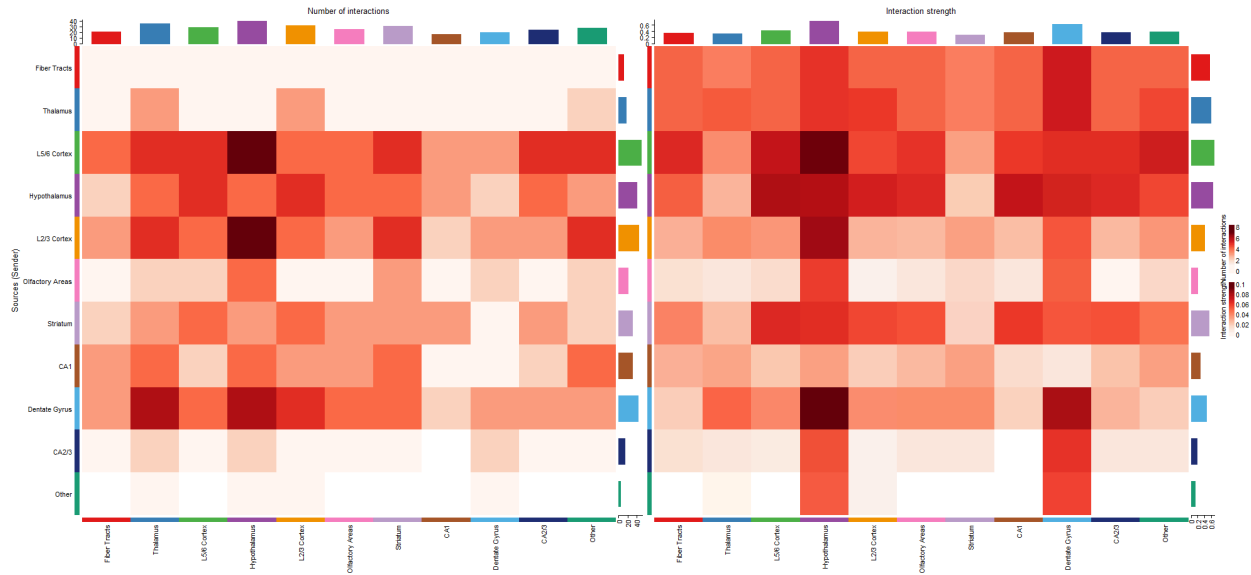


Fig. 15: Heatmap of cell-cell communication after processing WT 20mo through CellChat

Citations

Ji, J. X., Giles, B. L., Bhattacharjee, S., Kautzmann, M.-A. I., P. Masson, A., Do Carmo, S., Cuello, A. C., & Bazan, N. G. (2025). Intercellular Signaling and Synaptic Deconstruction Uncovered by Single-cell and Spatial Transcriptomics in an AD Tauopathy Model. *Communications Biology*, 8, 1583.
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